Statistics II: Introduction to Inference

Practice Problems (Set 1)

1. Let $F_{\boldsymbol{\theta}}$ be some distribution with mean μ and variance σ^2 (i.e., $E(X_1) = \mu$ and $\text{var}(X_1) = \sigma^2$), and $\boldsymbol{\theta} = (\mu, \sigma^2)$. Find the BLUE for μ.

2. Let the random variable X have pdf
$$f(x) = \sqrt{\frac{2}{\pi}} \exp\{-x^2/2\}, \qquad x > 0.$$
   (a) Find $E(X)$ and $\text{var}(X)$.
   (b) Find an appropriate transformation $Y = g(X)$ and α, β, so that $Y \sim \texttt{Gamma}(\alpha, \beta)$.

3. Let $X \sim \texttt{normal}(0, 1)$. Define $Y = -X\mathbb{I}(|X| \leq 1) + X\mathbb{I}(|X| > 1)$. Find the distribution of Y.

4. Let $X \sim \texttt{normal}(0, 1)$. Define $Y = \text{sign}(X)$ and $Z = |X|$. Here $\text{sign}(\cdot)$ is a $\mathbb{R} \to \{0, 1\}$ function such that $\text{sign}(a) = 1$ if $a \geq 0$, and $\text{sign}(a) = -1$ otherwise.
   (a) Find the marginal distributions of Y and Z.
   (b) Find the joint CDF of (Y, Z). Hence or otherwise prove that Y and Z are independently distributed.

5. Suppose $X_1, \cdots, X_n \overset{\text{IID}}{\sim} \texttt{normal}\left(\mu_x, \sigma^2\right)$, $Y_1, \cdots, Y_m \overset{\text{IID}}{\sim} \texttt{normal}\left(\mu_y, \sigma^2\right)$, and all the random variables $\{X_1, \cdots, X_n, Y_1, \cdots, Y_m\}$ are mutually independent. Then find the distribution of $T := S_X^{\star 2}/S_Y^{\star 2}$, where $S_X^{\star 2}$ and $S_Y^{\star 2}$ are the unbiased sample variances of X and Y, respectively.

6. Let $X_1, \cdots, X_n$ be iid random variables with continuous CDF F_X, and suppose $E(X_1) = \mu$. Define the random variables $Y_1, \cdots, Y_n$ as follows:
$$Y_i = \begin{cases} 1 & \text{if } X_i > \mu \\ 0 & \text{otherwise.} \end{cases}$$
   (a) Find $E(Y_1)$.
   (b) Find the distribution of $\sum_{i=1}^n Y_i$.

7. Let $X_1, \cdots, X_n \overset{\text{IID}}{\sim} \texttt{normal}\left(\mu, \sigma^2\right)$, and S_n^2 be the sample variance. Find a function of S_n^2, say $g\left(S_n^2\right)$, which satisfies $E\left[g\left(S_n^2\right)\right] = \sigma$.